

Data Quality Assessment Report

1. Datasets Overview:

The project utilized two raw datasets: `afl_players_info_raw.csv` (containing baseline player attributes) and `afl_players_seasonal_stats_raw.csv` (containing yearly performance metrics).

2. Dataset Profiles & Initial Findings

Dataset A: Player Information (`player_info_df`)

- Initial Dimensions:** 2,848 rows × 16 columns
- Missing Values:** High missing volume identified in non-essential text/URL fields: `profile_pic` (2,211 nulls) and `player_common_names` (2,773 nulls). Minor gaps in `player_teams` (94 nulls).
- Data Types:** Chronological attributes (`born_date`, `debut_date`, `last_date`) were initially stored as string objects.
- Duplicates:** 5 duplicate records detected and isolated.

Dataset B: Seasonal Statistics (`player_seasonal_stats_df`)

- Initial Dimensions:** 25,491 rows × 54 columns
- Missing Values:** 45 float-type performance metric columns contained partial missing values (totaling 10,026 affected rows across metrics like goals, tackles, and votes).
- Duplicates:** 10 duplicate records detected.

3. Data Cleaning & Resolution Log

Dataset	Identified Issue	Resolution Applied	Rationale
Player info	Massive missing URLs/names	Dropped columns('profile_pic', 'player_common_names')	Non-essential metadata/URLs that do not impact performance analysis.
Player Info	Missing team strings (94 rows)	Imputed with 'Unknown'	This preserves baseline demographic counts without deleting valuable player records..
Player Info	Dates stored as strings	Converted via <code>pd.to_datetime</code>	Standardizing date formats enables future temporal filtering, accurate career length calculations, and age-based validation.
Seasonal Stats	Missing Performance metrics (45 columns)	Imputed with 0.0	In sports performance metrics, missing numeric data points represent zero recorded instances during those game blocks
Both Datasets	Duplicate rows present	Dropped via <code>.drop_duplicates()</code>	Prevents distorted entity counting and false keys during dataset joining.

Seasonal Stats	Inconsistent capitalization & trailing spaces in team names	Standardized via <code>.str.strip().str.lower()</code>	Prevents duplicate categories and overlapping bars caused by case sensitivity (e.g., "Collingwood" vs "collingwood") during exploratory data analysis and visualization.
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4. Validation Summary & Post-Merge Metrics

A. Row Counts Before and After Cleaning

Player Info Dataset:

- Initial Shape: (2848, 16)
- Final Shape: (2843, 14) *(After dropping unessential columns and removing 5 duplicate rows)*

Seasonal Stats Dataset:

- Initial Shape: (15880, 54)
- Final Shape: (15874, 54) *(After removing 6 duplicate records)*

Merged Dataset (merged_players.csv):

- Final Shape: (15523, 68) *(Synthesized using an inner join on player IDs)*

B. Missing Values Handled

- **Player Info:** Minor missing values in `player_teams` were imputed with 'Unknown' to preserve baseline demographic counts, and unessential URL columns (`profile_pic`, `player_common_names`) were dropped. Date strings were standardized to datetime objects.
- **Seasonal Stats:** All numerical/float statistical missing metrics were filled with 0.0, under the rationale that unrecorded performance values during active match blocks logically represent zero occurrences.
- **Total Null Values Post-Cleaning:** 0

C. Duplicate Records Removed

- Identical duplicate rows were successfully dropped from both datasets to protect aggregate performance statistics and entity counts from distortion.

D. Unmatched `player_id` Values Identified After Merging

- **Orphan Seasonal Stats (Unmatched IDs):** 222 records
- **Profiles with No Seasonal Stats:** 1,068 records

Observations & Insights from the Cleaned Dataset

- **Data Integrity & Completeness:** Post-cleaning, the merged dataset achieved complete data integrity with zero null values (0), making it entirely stable for advanced aggregation, predictive modeling, or exploratory data analysis.
- **Dataset Scope & Join Dynamics:** Out of the total player profiles, a significant portion (1,068 profiles) lack seasonal performance logs, which could represent retired players, historical records prior to detailed tracking, or administrative entries. Conversely, 222 seasonal records lack corresponding player biographical profiles, indicating potential gaps in source registry maintenance.
- **Performance Distribution Readiness:** Filling numerical gaps with 0.0 ensures that player activity metrics (such as scoring, kicking, and tackling totals) can be accurately analyzed without dropping rows, preserving the full historical distribution of seasonal player output.